Yongye(Felix) Hu

510-277-7666 | hyy.felix@gmail.com | felix-hyy | felix-hu.space

EDUCATION

University of California, Berkeley — **Aug 2024 - May 2025**
Master of Engineering | Robotics Engineering — *Berkeley, CA*

Wuhan University of Science and Technology — **Sep 2020 - May 2024**
Bachelor of Engineering (Honours) | Mechatronics Engineering — *Wuhan, China*

WORK EXPERIENCE

Wuhan Xiaoxian AI Technology Co., Ltd. — **Sep 2023 - Mar 2024**
Mechatronics Engineer Intern — *Wuhan, China*

- Developed control programs for a six-axis CRP2 robotic arm using embedded C and ladder logic, enabling automated multi-axis motion for fuel rod testing.
- Designed an STM32F103-based gateway to enable Modbus RTU communication between the PC and robotic controller, ensuring stable and real-time data exchange.
- Proposed and implemented a magnetically-triggered encoder-based imaging system to replace low-precision pulse triggering, ensuring $\pm$1mm positional accuracy and reducing point-cloud data loss by 30%.
- Integrated IR, microswitches, and pressure transducers to monitor system state and trigger safety alarms, enhancing operational reliability and safety.

Lightweight, Sustainable Remote-Mining Excavator System — **Jan 2021 - Jul 2023**
R&D Engineer — *Wuhan, China*

- Designed a compact, remote-operated crane system with ESP32-based Wi-Fi control, enabling low-latency communication and autonomous coordination across multiple devices, achieving over 95% task repeatability.
- Enhanced the crane structure in SolidWorks with FEA; independently constructed STM32-based control circuitry and PCB with a stable fuzzy PID control system.
- Consolidated five functional boards into a single 4-layer PCB with KiCad, integrating control, power, and communication modules; reduced overall footprint by 40% and verified signal integrity via 100-MHz oscilloscope.

PROFESSIONAL EXPERIENCE

L.I.F.T: A Hybrid Flying and Driving Robot Platform — **Sep 2024 - May 2025**
Robotics Engineer & Project Status Manager — *Berkeley, CA*

- Led a 5-member cross-functional team to develop a hybrid flying-driving robot platform, achieving 100% integration of mobility and flight subsystems.
- Collaboratively designed a modular foldable robotic arm in Autodesk Fusion, reducing UAV-to-GAV transformation time by 50%.
- Integrated OptiTrack motion capture system with the robotic platform, achieving $<$1 mm position accuracy and $<$10 ms latency, supporting real-time position feedback for closed-loop control.

VacDry: Low-Temperature Vacuum Clothes Dryer — **Oct 2024 - Present**
Co-Founder & Product Engineer — *Berkeley, CA*

- Spearheaded mechanical design and development of an innovative vacuum dryer system from concept to working prototype, enabling 20-minute drying at 40°C with 50 dBA noise level.
- Performed FMEA on the pump, heater, and motion systems under simulated drying cycles, and designed a sensor-driven control interface with built-in diagnostics to simplify operation and facilitate fault isolation.

SKILLS

Design & Simulation: SolidWorks, Autodesk Fusion, Inventor, Ansys, MATLAB, AutoCAD, KiCad
Hardware: STM32, ESP32, Raspberry Pi, Modbus RTU, PLC, 3D Printing, CNC, PCB Design
Programming: C++, Python, ROS2, Linux, Qt, Embedded C, LaTeX